MIL-C-63162(AR)
25 June 1982
Superseding
(see 6.5)

MILITARY SPECIFICATION

CAP, ROCKET-OFF
M753 8 INCH PROJECTILE

This specification is approved for use by the US Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, quality assurance provisions and the preparation for delivery criteria for rocket-off caps used on M753, 8 Inch Projectiles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-B-121	Barrier Material, Greaseproof, Waterproof, Flexible
MIL-A-48078	Ammunition, Standard Quality Assurance Provisions, General Specification for

Beneficial comments (recommendations, additions, deletions), and any pertinent data which may be of use in improving this document, should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAN-Q, Dover, New Jersey 07801, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426), appearing at the end of this document, or by letter.

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STANDARDS

MILITARY

MIL-STD-1169 Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents form a part of this specification to the extent specified herein.

DRAWINGS (see 6.6)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

9333806 Cap, Rocket Off

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Construction. The rocket-off caps shall comply with all requirements specified on drawing (dwg.) 9333806, with all requirements specified herein and with the requirements of all applicable specifications to the extent specified on the dwg. and herein.

3.2 Materials. The contractor shall provide objective evidence (see 6.4) that materials comply with applicable dwgs. and specifications.

3.3 First article inspection. This specification contains technical provisions for first article inspection (see 4.3). Requirements for submission of first article samples shall be as specified in the contract.

3.4 Workmanship. Parts and assemblies shall be fabricated and finished in a thoroughly workmanlike manner and shall be free of deep scratches, burrs and sharp edges. The

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parts and assemblies shall be clean and free of chips, dirt, grease, rust, corrosion or any other foreign matter. Cleaning methods shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents. Protective finishes shall completely cover the surfaces to be finished with no bare spots and shall be free of deep scratches, gouges or other finish defects exposing base material. Marking on items and packaging shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The inspections specified herein are classified as follows:

1. First article inspection (see 4.3)
2. Quality conformance inspection (see 4.4)

4.3 First article inspection.

4.3.1 Submission. When first article inspection is required (see 3.3), a first article sample shall be submitted as designated by the contracting officer for evaluation in accordance with 4.3.2. The first article sample shall consist of the following items in the sample quantities as indicated.

Part Description	Drawing	Quantity
Rocket-Off Cap	9333806	10

4.3.2 Inspections to be performed. The first article sample submitted in accordance with 4.3.1 shall be subjected to any or all of the requirements of the applicable dwg. and

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to the tests and inspections specified in the First Article Inspection Table herein. All parts shall have been manufactured using the same production processes, procedures and equipment which will be used in fulfilling the contract and all materials shall be obtained from the same source as will be used in regular production.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.3.2.1	Cap, Rocket-Off, M753 Projectile				9333806
					NEXT HIGHER ASSEMBLY
					M753 Projectile
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Classification of Defects and Tests	10		3.1, 3.2, 3.4	4.4.2.1

NOTES:

DRDAR-QA Form 160 Jul 77 Replaces SARPA-QA Form 2567 Feb 74 Which is Obsolete

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4.3.3 Inspection equipment. The inspections and tests specified in 4.3.2 shall be accomplished with the gages and test equipment specified in 4.4.4.

4.3.4 Rejection. See MIL-A-48078.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots of caps shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examinations. See MIL-A-48078 and the Quality Conformance Inspection Tables herein

(a) Major defects: Unless otherwise specified, inspection for Major defects shall be sampled in accordance with Table I.

(b) Minor defects: Unless otherwise specified, inspections for Minor defects shall be sampled in accordance with Table II.

TABLE I. Sampling plan for major defects.

LQ = .05 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

TABLE II. Sampling plan for minor defects.

LQ = .10 Acceptance No. = 0 Consumers Risk = .10

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.1	Cap, Rocket-Off, M753 Projectile				9333806
					NEXT HIGHER ASSEMBLY
					M753 Projectile
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	1.018 outside diameter (2 places)		100%	3.1	Gage
102.	Major diameter of thread		100%	3.1	Gage
103.	Pitch diameter of thread		100%	3.1	Gage
104.	True position of major diameter of thread		100%	3.1	Gage
105.	Outside diameter of O-ring groove		100%	3.1	Gage
106.	True position of O-ring groove		100%	3.1	Gage
107.	Width of O-ring groove		100%	3.1	Gage
108.	63 surface finish (4 places)		100%	3.1	Visual
109.	Small outside diameter (boss)		100%	3.1	Gage
110.	Width of small outside diameter (boss)		100%	3.1	Gage
111.	Width of flat on boss		100%	3.1	Gage
112.	Taper on boss		100%	3.1	Gage
113.	True position of boss		100%	3.1	Gage
114.	Corner radius in O-ring groove (2 places)		100%	3.1	Gage
115.	Outside diameter of hub		Table I	3.1	Gage
116.	Overall length		Table I	3.1	Gage
117.	Location of O-ring groove		Table I	3.1	Gage
118.	Length of undercut		Table I	3.1	Gage
119.	Outside diameter of undercut		Table I	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1 (cont)	Cap, Rocket-Off, M753 Projectile	2 OF 2	9333806
			NEXT HIGHER ASSEMBLY: M753 Projectile

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
120.	Face to bottom of cavity		Table I	3.1	Gage
121.	Location of O-ring groove from boss		Table I	3.1	Gage
122.	Chamfer on face of thread		Table I	3.1	Gage
123.	Inside diameter of cavity		Table I	3.1	Gage
124.	Taper in cavity		Table I	3.1	Gage
125.	Radius inside of cavity		Table I	3.1	Gage
126.	Thickness of tabs		Table I	3.1	Gage
127.	Width of tabs (3 places)		Table I	3.1	Gage
128.	Center to end of tab (3 places)		Table I	3.1	Gage
129.	Location of tabs (3 places)		Table I	3.1	Gage
130.	Radius hub to tab (6 places)		Table I	3.1	Gage
131.	Radius on end of tabs (6 places)		Table I	3.1	Gage
132.	Part number or marking improper or missing		100%	3.1	Visual
133.	Painted dots (3) improper or missing		100%	3.1	Visual
134.	Protective finish improper or missing		100%	3.1	Visual
135.	Material		-	3.2	Visual (6.4)
136.	Evidence of poor workmanship		100%	3.4	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Packaging				See 5.0
					NEXT HIGHER ASSEMBLY
					N.A.
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
101.	Packaging and packing not in accordance with applicable requirements		-	5.0	Visual
102.	Objective evidence (see 3.2) not available for packaging and packing material		-	5.1.1	Visual
103.	Parts not clean and free of dirt and other foreign material prior to packaging		100%	5.1.1	Visual
104.	Parts not individually packaged		100%	5.1.1	Visual
105.	Stiffening and cushioning of packages inadequate to provide firm package (when required)		100%	5.1.1	Visual/Manual
106.	Individual packages not properly heat sealed		100%	5.1.1	Visual
107.	Intermediate pack improper or not properly sealed		100%	5.1.2	Visual/Manual
108.	Exterior pack improper or not properly sealed		100%	5.1.3	Visual
109.	Marking of individual packages, intermediate packs or exterior packs missing, improper or illegible		100%	5.1.4	Visual
110.	Evidence of poor workmanship		100%	-	Visual

NOTES:

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the table starting with 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract (see section 6 of MIL-A-48078 and 6.3 herein).

4.5 Test methods. Not applicable.

5. PACKAGING

5.1 Packaging and packing. Packaging and packing shall be in accordance with MIL-STD-1169 subject to the following details and conditions.

5.1.1 Unit Packaging. Items shall be individually packaged in fiberboard containers, sealed in waterproof barrier bags and heat sealed in accordance with MIL-STD-1169, Level A. Barrier bags shall be fabricated from MIL-B-121 (Type II, Grade A, Class 1) material or an approved equivalent. Fillers, positioners, stiffeners and cushioning adequate to prevent physical damage to the item and to finished surfaces during transit, storage and handling shall be provided as required.

5.1.2 Intermediate Packaging. Unit packages shall be packaged in intermediate packages in accordance with MIL-STD-1169, Level B.

5.1.3 Packing. Intermediate packages shall be packed in accordance with MIL-STD-1169, Level B.

5.1.4 Marking. Unit packages, intermediate packages and exterior containers shall be marked with the following:

a. Item nomenclature

b. Part or dwg. no.

c. Used on M753 Projectile

d. Manufacturer

e. Quantity enclosed

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f. Lot no.

g. Contract or purchase order no.

h. Date of acceptance

6. NOTES

6.1 Intended use. Rocket-off caps covered by this specification are intended for use on Rocket Motor/Bulkhead Assemblies for M753, 8 Inch Projectiles.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this inspection.

b. First article inspection, when required (see 3.3).

6.2.2 Data requirement. When this specification is used in an acquisition which incorporates a DD1423, Contract Data Requirement List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirements	Applicable DID no.
(a) 4.4.4	Inspection equipment approval	DI-R-1714 (Tailored)

6.3 Equipment design approval. Submit equipment designs, as required, to Commander, U.S. Army Armament Research and Development Command, ATTN: DRDAR-QAN-I, Dover, New Jersey 07801.

6.4 Objective evidence. A statement (certification) supported by test data that each lot of material meets the requirements of the applicable dwgs. and specifications.

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6.5 Supersession data. This specification includes the requirements of Purchase Description AR-PD-29 dated 14 November 1979.

6.6 Drawings and specifications. Drawings and specifications listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command may also include dwgs. and specifications prepared and identified as Frankford Arsenal, Edgewood Arsenal, Rock Island Arsenal or Picatinny Arsenal dwgs. and specifications. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:

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Preparing activity:

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-63162 CAP, ROCKET-OFF M753 8 INCH PROJECTILE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

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B. RECOMMENDED WORDING CHANGE

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